

Airbnb-Style Accommodation Booking System

GROUP 89

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Task 1: Project Scope and Business Requirements

1.1 Introduction

The Airbnb-style accommodation Booking System is an online platform designed to facilitate short-term accommodation bookings. The system connects guests looking for temporary stays with hosts who list their properties. The primary goal of the project is to design and implement a reliable Database Management System (DBMS) that efficiently manages users, properties, bookings, payments, and reviews.

The system ensures accurate booking management, prevents conflicts such as double booking, and maintains data integrity while supporting concurrent user access.

1.2 Scope of the System

The scope of this project is limited to the database design and basic application interaction required to demonstrate core DBMS concepts.

The system will:

- Manage user accounts for guests and hosts

- Store and manage property listings
- Handle property availability and bookings
- Record payments and reviews
- Support basic administrative monitoring

1.3 Stakeholders

- **Guest**
 - Registers and logs in
 - Searches and books properties
 - Makes payments
 - Cancels bookings
 - Submits reviews
- **Host**
 - Registers and logs in
 - Lists and manages properties
 - Sets availability and pricing
 - Views bookings
- **Administrator**
 - Monitors users, properties, and bookings
 - Views system statistics and revenue

1.4 Functional Requirements

1. Users must be able to register using a unique email and phone number.
2. Hosts can create and manage property listings.
3. Guests can search properties based on location and availability.
4. Guests can book available properties for selected dates.
5. The system must prevent overlapping bookings for the same property.
6. Payments must be recorded for each confirmed booking.
7. Guests can cancel bookings according to system rules.
8. Reviews can be submitted only after a completed booking.

1.5 Business Rules and Constraints

- Each property is owned by exactly one host.
- A booking is associated with one guest and one property.
- A booking cannot exist without availability.
- Each payment corresponds to one booking.
- A review can only be submitted for a completed booking.

1.6 Non-Functional Requirements

- **Scalability:** The database should handle multiple users and bookings.
- **Data Integrity:** Enforced using primary keys, foreign keys, and constraints.
- **Concurrency Control:** The system should handle simultaneous booking attempts.
- **Security:** User credentials and sensitive data must be protected.

1.7 Assumptions

- Payment processing is simulated.
- The system supports a limited number of users for demonstration purposes.
- The UI will be basic and used solely to demonstrate DB operations.

2. Technical Requirements

- MyS
- QL
- Python
- HTML
- CSS
- JavaScript
- ReactJS
- Flask

3. Detailed Overview of the Functionality

3.1 Guest Management

- **Signup:** Account creation with unique email and phone number and secure password constraints
- **Login:** Authentication using registered credentials
- **Property Listings:** Searchable property catalogue
- **Search & Filter:** Find properties by location, price, and availability
- **Booking Cart:** Temporary storage of selected bookings
- **Payment:** Secure booking payment with transaction history
- **Booking Tracking:** View booking status and stay details
- **Booking History:** Access past bookings and reviews
- **Sign Out:** Session management

3.2 Host Management

- Signup and Login with unique credentials
- Property listing management
- Availability and pricing management
- Booking management
- Earnings tracking

3.3 Admin Management

- Secure admin login
- User and property management
- Booking trend analysis
- Revenue statistics